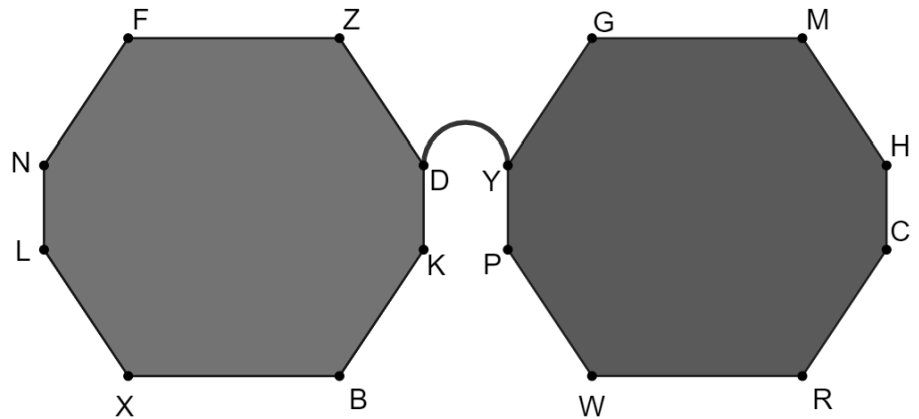


Geometry
Unit 5
Lesson 10 Homework

Name _____
Date _____
Period _____

Minnie is designing sunglasses. She uses a computer drawing program to create a diagram of the lenses. Her drawing is shown with congruent polygons $FZDKBXLN$ and $GMHCRWPY$ representing the lenses of the sunglasses.



In polygon $FZDKBXLN$, $\overline{FZ} \cong \overline{XB}$, $\overline{NL} \cong \overline{DK}$, and $\overline{DZ} \cong \overline{KB} \cong \overline{LX} \cong \overline{NF}$. The perimeter of polygon $GMHCRWPY$ is 56.8 millimeters (mm).

Some of the side measures of $GMHCRWPY$ are shown.

$$PW = 6x - 2.4$$

$$HC = 2(3x - 2.8)$$

$$WR = 5x + 2$$

- Write an equation that can be used to determine the value of x .
- Determine the value of x .
- Determine the length of side GM .
- Determine the length of side BK .
- Determine the length of side NL .

Some of the angle measures of $FZDKBXLN$ are shown.

$$m\angle BXL = (4k - 8)^\circ$$

$$m\angle DZF = (3k + 25)^\circ$$

$$m\angle XLN = (5k - 19)^\circ$$

$$m\angle XBK = (5k - 41)^\circ$$

$$m\angle NFZ = m\angle DZF$$

$$m\angle DKB = (3k + 47)^\circ$$

$$m\angle XLN = m\angle LNF$$

$$m\angle ZDK = m\angle DZF + 22^\circ$$

6. Given that $m\angle NFZ + m\angle FZD + m\angle MHC = 394^\circ$, determine the value of k .

Determine the measure of each angle.

7. $m\angle BXL = \underline{\hspace{2cm}}$

8. $m\angle WRC = \underline{\hspace{2cm}}$

9. $m\angle PYG = \underline{\hspace{2cm}}$

10. $m\angle RCH + m\angle CHM = \underline{\hspace{2cm}}$